Nikhil Kamath ft. Perplexity CEO, Aravind Srinivas | WTF Online Ep 1.

[Speaker 2] (0:00 - 0:03)

Are you like a Chennai boy? Have you grown up there all your life?

[Speaker 1] (0:03 - 0:06)

My parents live in Chennai, so I first go there.

[Speaker 2] (0:06 - 0:07)

What were these fancy ideas?

[Speaker 1] (0:08 - 0:12)

I'm very upset to hear that, because I actually thought my ideas are cool.

[Speaker 2] (0:45 - 0:57)

Hi Arvind. Hi Nikhil. Hi.

This is a bit weird for me, because I'm doing it this way, a conversation after a bit. Okay. Yeah.

[Speaker 1] (0:58 - 1:04)

Yeah, I wish we could be in the same place. Where are you now? I'm in San Francisco.

[Speaker 2] (1:05 - 1:06)

Right.

[Speaker 1] (1:06 - 1:16)

Yeah, I was traveling to Europe last week. Okay. A lot more travel coming up, but hope to be in India pretty soon, before my May, hopefully.

[Speaker 2] (1:18 - 1:22)

That's not far. When you come to India, where do you typically go to? Is it?

[Speaker 1] (1:23 - 1:50)

My parents live in Chennai, so I first go there. And then, depending on the arrangement, like who I'm meeting. I spent the last time I came, I went to Mumbai and Delhi.

And this time, I probably will try to go to Bangalore too, in addition to these two cities. But everything's like in the flux right now.

[Speaker 2] (1:51 - 1:56)

Right. Super. Are you like a Chennai boy?

Have you grown up there all your life?

[Speaker 1] (1:57 - 1:57)

Yeah.

[Speaker 2] (1:58 - 2:03)

So, you know, like, the local stuff about Chennai kind of thing.

[Speaker 1] (2:05 - 2:14)

I mean, I'm, yeah, I grew up there. So I hope, I don't know exactly what you mean by local, but I definitely know Chennai pretty well.

[Speaker 2] (2:15 - 2:27)

Right. How did this begin? Would you like to start by telling us like a little bit of, little bit about your journey, how it was from where you began in Chennai to where you are today?

[Speaker 1] (2:29 - 7:20)

Yeah. Well, I was just like any other student in Chennai, just studying. People in Chennai study a lot.

I think that's one thing I've known. I think I was pretty interested in all sorts of statistical things, mainly coming from following cricket a lot. As people generally like try to analyze the stats and run rate and like how many 50s, 100s.

And I got an intuitive sense for numbers pretty early on. I was pretty good at math. And I also like early on picked up programming towards the end of my, I think 11th standard.

So that was how it began. And obviously my mom wanted me to get into IITs. Every time like we would go on a bus and pass by the IIT Madras campus, my mom would point to the campus and say, this is where you're going to study.

It's not even like you should study, this is where you're going to study. So that was the expectation. I definitely grew up thinking that, okay, I do want to like compete and win against the best people.

And it was the JEE exams are pretty competitive, as you know. And so we had a pretty good rivalry among like fellow students. And I obviously didn't do as well as I wanted in the JEE, but I got into IIT, got into electrical engineering.

And inside the campus, again, like a lot of our friends got into competitive programming. So I was into that too. But I figure I was not as good as what you needed to be to get to the world finals of like ICPC or something.

And so I got a good understanding of computer science and got a lucky opportunity to learn about machine learning pretty early on. I got a roommate of mine, or like someone neighboring to my room, told me about this contest that was running in Kaggle, where you just had a data set and you had to predict stuff. And I had no clue what any of these meant.

They were just a bunch of numbers you downloaded and figured out a prediction classifier for like unseen outputs. And that's when I got into things like Scikit-learn, which was a very famous machine learning library. And it just randomly tried all these algorithms, mix and match them.

That got me, that helped me win the contest. And then from there onwards, I thought, OK, like maybe I should take this more seriously. I got an internship in a startup at Bangalore and built recommender systems for them pretty quickly, actually.

So my internship, which is supposed to be two and a half months, I finished it in like three weeks. They submitted my solution to a client, got the money. So they were happy.

And so I got a lot of time to just sit in, like learn ML, but come to the intern office and just learn. So that's what I did. I self-taught myself all the Andrew Ng lectures and did all the Stanford materials, went back to campus, took the machine learning class there, started doing research that got me a PhD in Berkeley.

And then like from there, I got an internship at OpenAI DeepMind, built my fundamentals. And obviously, every stage I got better and better peer group exposure, always questioned whether my understanding of the world was correct or not. I felt comfortable being like not the best person in the room.

That takes time because the IAT mindset is like, I want to be the best, smartest person in the room. I think after coming here, I learned it's OK to not be the smartest person. It's OK to be the person that wants to learn everything and learn from the best.

And I think that was very formative once I came here. Because when I actually went for my internship at OpenAI, it was truly humbling. I was very, very bad compared to the people there.

And I thought I was good. So those were the years when I really learned the details of AI and machine learning. And I think it's really helped me to come to where I am right now.

[Speaker 2] (7:21 - 7:23)

What years were these, Arvind, at OpenAI?

[Speaker 1] (7:24 - 10:01)

So my summer internship was in 2018. So the way it happened is I came to Berkeley, I didn't have an advisor. So, you know, when you don't have an advisor, they give you like a lab space that was very small, like somewhere in the corner.

And so that was not very stimulating to go and work every day. But again, I'm not a person who tries to blame things. So I just go to the Philz Coffee here every day morning.

I would go wake up at 530 a.m., I would be there, I would be the first person at Philz Coffee, and I would leave at 8 p.m. in the evening. And I would just work all, like every single hour, I would just work. You know, because I didn't have my own computer, like to do my research, I learned how to use the cloud and just work from my laptop.

And all these things were helpful. And I wrote a paper pretty quickly, and that got me my advisor, Peter Abbeel. And his student was John Shulman, who was one of the OpenAI co-founders, and he famously went on to create ChatGPT.

So that guy invited me because he noticed my work, and we were all from the same lab, so he noticed my work and he invited me for an internship. And that's when I went to OpenAI, and like Ilya Sutskever was essentially running the company at the time. There was no Sam Altman, or I think Elon Musk was on his way out too.

So Ilya listens to me for like half a minute about my ideas, and he just says, you're wrong, like all your ideas are useless. And not in a way where it's like arrogant, he's just like literally just telling the truth in a very respectful way. And I'm very upset to hear that, because I actually thought my ideas are cool.

And that's what I was being told by other people on campus. So I then go on to question, and he just tells me AI is just two circles, and he draws a big circle, and then inside that he draws a smaller circle. And he said the big circle is generative AI, and the smaller circle is reinforcement learning, RL.

And together, this is the recipe for making AI happen, AGI. And the only thing that remains is throw a lot of compute at it. And he said this in 2018.

And I was doing very fancy ideas that, you know, were like made me feel smart, but not necessarily that mattered long run.

[Speaker 2] (10:02 - 10:04)

What were these fancy ideas?

[Speaker 1] (10:05 - 12:12)

I was trying to work on things where the AI would learn its own loss function. So obviously, the thing in AI is like there's something called the loss function. That's what the neural network optimizes, right?

So when you're trying to build intelligence, you don't actually know what is the real loss function, right? You could say intelligence emerges from predicting the next word from the previous word. But someone could say, why is that a sign of intelligence too?

You could say intelligence comes from identifying thousand breeds of cats and dogs, and like, you know, all the images, you just assign a label to it and predict the label. But you could say, okay, wait, humans don't learn like that. So there's no one magic sauce for building a generally intelligent model in AI.

You can design objective functions that are narrow in nature, like, oh, like you go and master the game of Go or chess or like, be the best object detector on the planet. But these are not going to lead to general intelligence. So I was trying to work on research where the AI comes up with its own loss function, trains on it, and then evaluates itself on a bunch of tasks, and then decides, okay, it has to go and tweak the loss function a little bit to be better at like more tasks.

And then I keep doing this iteratively. And then I thought that's in that loop, intelligence will emerge. This is a good idea, right?

But Ilya just said, this is too complicated. I think from the beginning, mainly my main takeaway from opening an internship was, like, even though other people in academia who are like the elites will respect you for the more complicated ideas, what matters in reality is making things work. And it's often the simplest ideas that work in practice, especially when thrown a lot of compute at them.

The simplest ideas typically outshine the complicated ones.

[Speaker 2] (12:13 - 12:35)

So when we talk about AI today, let me like set context. Think of me as an absolute idiot, who does not understand anything. And whenever you say something, if possible, please try and explain it to me in the manner that you would speak to like a 10 year old boy who's not very smart.

That would help.

[Speaker 1] (12:36 - 12:37)

Sure, absolutely. Yeah.

[Speaker 2] (12:39 - 13:39)

And I think a good place to start. Today, where I am, I work in FinTech largely in India, but I feel whenever I read the news, or I watch the news, very insecure about the fact that so much is happening in AI. And I almost feel like I'm being left out of it.

And it doesn't feel like I'm even amidst the action to learn about it, it feels like I'm talking to the commentator, or reading what the commentator who has what he has to say, whereas the match is happening in another region altogether. So maybe we can preface this conversation with like maybe a brief history of compute leading up to AI, in the manner that you would speak to a 10 year old boy. And we can take it from there.

[Speaker 1] (13:40 - 13:59)

Sure. I mean, AI has been going on since a long time. If you're like, in, I think, there was a project at MIT, which declared you can solve AI in a summer project, like, like, literally in three months.

And obviously...

[Speaker 2] (13:59 - 14:00)

Would you want to first define what is AI?

[Speaker 1] (14:01 - 14:10)

So AI, obviously, artificial intelligence is a field of computer science that's trying to design computers to behave intelligently.

[Speaker 2] (14:10 - 14:13)

I was wondering what their definition of intelligence is.

[Speaker 1] (14:13 - 14:39)

Yeah, yeah. Program computers to do tasks that require some level of intelligence to accomplish them in a manner similar to a human does it. And what is the scope of tasks that require intelligence that you want the computer could do, is where the generality comes in.

So...

[Speaker 2] (14:39 - 14:46)

So are you saying that intelligence is when a computer is able to behave like a human, because that itself...

[Speaker 1] (14:46 - 16:31)

General intelligences. General intelligences. Right.

So an AI that you write, you know, for a chess game that you're building, let's say you're building a chess game as a software project. And obviously the... when the user picks white, and the blacks playing with an AI, that is an AI you write for the game.

That is not really a generally intelligent AI. It can only do what you hard-coded it to do. It can assign points for each piece.

A bishop is this much, a knight is this much, and it can just run a tree search to optimize for that. By that, I mean, it can search for moves, roll out a few steps, and then try to pick the one that gives you the maximum score. That is...

people used to call that AI, but that is not general AI. The reason is, whatever software you write for that cannot do another game even, leave alone another task. It is a very constrained, specific setting.

Now that is interesting by itself. There are a lot of things you could do in the world that are useful where you break down a problem, and you write a specific solution for that. It is pretty useful.

But what was really on the frontier of science at that time when I was doing PhD was like, how can we figure out general intelligence in a manner similar to a human, which is one system doing hundreds of thousands of tasks without explicitly being programmed for it, and can be taught new tasks, and it can learn on the fly without much effort.

[Speaker 2] (16:32 - 16:47)

What is it optimizing towards though? Let's say AI or AGI is able to do millions of different tasks, learn along the way. If I were AGI, how would I decide what to do first?

[Speaker 1] (16:48 - 17:40)

Yeah, so I think that is where we get even further in terms of what is an AGI. Is this like an agent that is constantly deciding what to learn next on its own? Does it have autonomy?

Or is it still a generally intelligent software, but it does not have any autonomy of its own, and it does not decide what to do next of its own? Is it actually aware of its own limitations, and then it's deciding, okay, I lack this task, and this is what I need to go and learn next. No, that's not what we have today.

Ideally, we should, right? What you're suggesting is like an AI that not only learns and trains on stuff that the humans throw at it, but also decides what to do next in terms of how to make itself

better, recursive self-improvement. That is not correct yet.

[Speaker 2] (17:41 - 17:48)

So is the AI trying to make itself better? If AI can do anything, would it want to make itself better? Would that be the motive?

[Speaker 1] (17:52 - 20:39)

Like, that would be the ultimate motive. If you have an AI that constantly keeps trying to improve itself on any task that it wants to, and reasons that this is the thing that it's worth working on for the next step for itself, then I think that would be the ultimate version of, some people call it even superintelligence, which is AI has gone beyond the realms of AGI, which today's systems are, let's say, okay, for the sake of our discussion here, let's define AGI as like a very, very smart version of like, you know, current models, like say GPT 4.5 or 5, 2 or 3 generations later, 6 or 7, where they're doing most of the tasks that we do on a computer on their own pretty well.

With just a simple language instruction, they can just do it. I think that system, a lot of people might want to call that as an AGI, even though it's not, you know, completely an AGI. It's not doing physical work that a human does, and physical work also requires intelligence.

Let's just say that is a pretty reasonable working definition. Now, that system still doesn't have awareness of itself. It's not aware of what it's bad at, what it's good at, and like, what it should do next.

What are its real goals? It's not autonomous. It's not aware.

So, the ultimate problem in AI is like, how do you build an agent that does exactly whatever these models do, but also can keep improving itself and can keep coming up with its own goals? And what is its real true objective function? Is it to help humanity?

That's what people get spooked about, usually, when they talk about like AI taking over humans. Now, I think in AI community right now, people are typically referring to the second part as super intelligence, not general intelligence, where once you crack it, there is no way to like control it, because you can just say, until you shut down the system, it's just going to keep improving itself. But then there is an argument also that, okay, if it's that smart, that it knows what to do all the time, and it's thinking so smart ahead of everybody else, why would it not predict that humans might want to shut it down and create clones of itself and keep staying alive?

So, that's where like, those are getting more in the sci-fi territory. But whatever we are working with today is like, okay, it's some 10,000 knowledge worker professions in one system without any like hard coding. That's already pretty crazy.

[Speaker 2] (20:41 - 20:51)

Right. I'm still trying to wrap my head around the definition of intelligence almost being human like behavior. If that's what you mean.

[Speaker 1] (20:54 - 21:20)

I think like the definition that I would say is practically pretty useful right now is, can you create a digital remote knowledge worker? I think that is what like most people are working towards. It's kind of converged to that digital remote knowledge worker like an employee that you can hire on Upwork.

Can it just be an LLM?

[Speaker 2] (21:21 - 21:34)

But is that intelligence truly like if you're able to replicate some human abilities onto an agent? Would you would you consider that intelligence? Is that the definition that one goes with?

[Speaker 1] (21:35 - 24:40)

Well, I think like some people are pretty precise about what intelligence is. It's like until you get me the human brain equivalent in software, nothing is intelligent. Everything is like narrow.

And there is some merit to that argument. And it's pretty difficult to create something exactly like the human brain. Because the human brain is amazing.

It's power efficient. It doesn't consume all the data centers in the world to do the tasks that we do. And it's pretty fast at learning new things.

It does physical intelligent work to not just digital dexterity, all that stuff. So yes, you can argue it's not really intelligence in the human way. But the other way to look at it is the functional way.

Just look at the output and the input. I give the human the same output input and I give the AI the same input. And does the AI work better than a human on tasks that humans are actually getting paid for in the world?

Software engineering, one of the highest paid professions. Today, it's pretty obvious that most human software engineers, at least like the median human software engineer is probably worse than an AI today. Typically, we have considered people who write code as smart people.

It's just like thing we've done. And so now when an AI is able to do that, if we say that is not intelligence, then it's kind of like also saying, okay, humans writing code was never an intelligent thing either. You got to apply the same standards.

So then what is intelligence kind of changes to? It's not the fact that humans wrote code that made them intelligent. It's the fact that they can do writing code and designing art and building a

home.

All these things in one person, that is intelligence. Now, one could again argue that there are very few people who are good at doing all these three simultaneously. Most people are good at only one or two things and they have hobbies, but they're never world-class at the hobbies.

And so that's where we are getting at. If humans are also limited in what they can really do in a world-class manner, and one AI system that can write code better than the median software engineer is also writing emails better than the median executive assistant, and is also doing writing essays better than the median writer, that's pretty intelligent. That system or whatever software system, it's definitely not human-like, but as an output that it produces, it's pretty good or better than most humans right now.

And so I would consider it an intelligent system, right? And very different kind of intelligence than like a calculator or a chess program.

[Speaker 2] (24:41 - 25:06)

Yeah, no, I actually like the definition of intelligence that if a computer is able to mimic what a human does or can do, is intelligence. But then along that spectrum, computers for a long time have been able to replicate or better human tasks, different things, like maybe mathematics, for example. Has all of that been under this purview of intelligence?

[Speaker 1] (25:08 - 25:42)

They used to be done in the research field of AI, like when Deep Blue beat Kasparov in chess, it was considered AI research, Monte Carlo research was AI research, but then people were not like, oh, this is insane. People were more like, how do we make this work in a more general way? That it's not just doing it for one task.

How do we do something?

[Speaker 2] (25:42 - 25:50)

Does intelligence have to be general? It can be narrow, and if something is intelligent at one thing, you would still define it to be intelligent, right?

[Speaker 1] (25:50 - 25:54)

Yeah, sure. I think you can definitely define it to be intelligent.

[Speaker 2] (25:55 - 26:06)

So what I'm kind of looking for is the distinction between a calculator doing maths and what we call intelligence today, because a calculator does something a human could do better than he

could do.

[Speaker 1] (26:06 - 27:42)

I think you can definitely call it intelligence. But then that's when you call anything an AI, like anything that makes any kind of prediction becomes an AI. The reason that people genuinely consider a more general system is more intelligent is because it's harder to just overfit a solution to 10,000 problems at once.

It's not like 10,000 programs being written and then stitched together. It's more like one program that's able to do equivalent to 10,000 programs simultaneously. The same system, the same piece of software, which is just a bunch of weights of a neural network, whatever input you feed to it, you ask it to write code, you ask it to write a poem, you ask it to write an essay, you ask it to summarize a document.

The same system does these tasks in one way, right? I think that is what is amazing. That's the generality coming from.

I could have written 10,000 different programs each for each of these different tasks and have a router that tells which program to use for what input you will give me. That would still appear intelligent to you, but it's not truly general. When you throw the 10,001 task, slight variation, I might not be able to do it.

But a more general system that just uses one piece of code will be able to do it. I think that is the power of generality, like the fact that humans can keep...

[Speaker 2] (27:42 - 28:09)

In a way, you're also saying that intelligence in today's context, we're all talking about AI having changed drastically over the last, I don't know, whatever time frame. So you're saying it has moved from narrow to more general in nature where it can't just do one task like a calculator, but it can learn how to do another and solve it? Correct, yeah.

Is that the distinction you're drawing?

[Speaker 1] (28:10 - 29:03)

I think that is why people are way more excited. And interestingly, it is able to do stuff that people are getting paid for. So it'll have economic implications.

Unlike the previous cases where it mastered chess or Go, people found it cool, but nobody really cared because they couldn't use it on their own on a daily basis unless you're a professional player. On the other hand, a lot of people are writing code, a lot of people are writing documents, a lot of people are summarizing things. I think that's where I would say it's beginning to feel like

you hired another human for your work.

Right, right. And so it's truly replacing human labor in a meaningful way.

[Speaker 2] (29:04 - 29:31)

Yeah, this is another stupid question, but I'm going to ask it. How a calculator worked. I'm using the most basic of examples, how a calculator worked back in the day.

When I hit on a computer, multiply 25 into 25 or 20 into 20, what happened on the backend that I couldn't see? How did it throw the output? We'll start there and we'll try to extrapolate all the way to what's happening today.

[Speaker 1] (29:31 - 30:44)

Sure. I mean, there are circuits for adders, multipliers, and these are the circuits that are running on the back. Depending on your input that you enter, it's getting parsed, and then that input is getting fed into the circuits, and then you're getting the output.

And you can build even mechanical circuits. That's a beautiful thing. Once it works, there are nice visualizations of how adders work in a completely mechanical way.

So you don't actually need that much power to make this work. I saw somebody say a beautiful thing. A calculator is such an amazing artifact that if you took it, let's say from 2025, and let's say your time traveled back to 1800, it would still work the same way.

Let's say it's solar powered, it would just work the exact same way. And that's fantastic because you cannot say that about, let's say, a MacBook. You're not going to be able to power it, right?

[Speaker 2] (30:45 - 30:57)

That's actually useful. The way you described it helps me visualize. When you say it's mechanical, I can imagine an IC doing what I can picturize mechanically in my head.

[Speaker 1] (30:57 - 31:16)

Yeah, there are some. You can go to YouTube and watch how you can have a binary counter, three-digit binary counter that's completely mechanical. And it's pretty beautiful.

Okay.

[Speaker 2] (31:17 - 31:19)

And then what changed in computing after that?

[Speaker 1] (31:20 - 31:57)

Well, a lot of things, obviously. We've got to go very deep into what people built with calculators, other devices and so on. But I think the biggest change, I would say, was the personal computer revolution.

We had mainframe computers, right? But then the biggest change that truly made computing very democratized and ubiquitous is people being able to have a personal computer at home. That was the whole Apple 1, Apple 2, IBM.

Yeah.

[Speaker 2] (31:57 - 32:09)

So sticking to the IC example, Moore's Law, ICs got smaller and smaller so you could have enough compute at home to do these same calculations that before you needed a mainframe. Correct. Correct.

[Speaker 1] (32:09 - 33:38)

Yeah, definitely Moore's Law is one of the critical reasons it happened. But also a lot of artistry in the beginning to package a lot of computations in a very compact way into one board that could be put into a portable computer was pretty amazing. A lot of people actually in the beginning were skeptical.

They thought that it's not going to matter. Why would need a computer at home, right? It's just like stuff you do for work.

And that's where the beauty was, hey, people might want to actually work at home too. Of course, games was a big deal, but the real reason personal computers took off was the software called VisiCalc, which is essentially a spreadsheet and calculator. And so that led people who were doing accounting to do the work at home.

And slowly it spread. More software started being written for personal computers. And so that made personal computing fun.

Now, after that is the network effects where if you had a personal computer at home and I had one, and we could figure out a way to talk to each other, which is internet, and then the worldwide web, and then like mobile, cloud, and now AI. So it's very simplistic way to describe it. And there's a lot of details here,

[Speaker 2] (33:38 - 34:52)

but this is actually very useful because I feel like whenever I try and learn more about this field online, or with the people I speak to, I feel like I get the high level, non nuanced, generic stuff that everybody's saying, but I'm not able to, I don't have that bridge in my brain, which goes from

Okay, it started like this, then it just happened, then that happened. I did.

I interviewed Jan recently on Lacuna a few months ago, and yeah, he's awesome. And we went into like, I spent a lot of time sitting and trying to learn about Jepa and the machine learning and neural networks and what his creations were. But again, the manner in which I explained it, or we kind of like try to portray it, I think got too muddled, because I don't have a clear understanding of much of this.

So let's say when you say we moved from the internet to like today's AI, when everybody's talking about AI, what was that one thing? If you had to fixate on one thing, why is AI in 2025 different from when people spoke about AI in 2010?

[Speaker 1] (34:53 - 35:52)

I think the biggest change from 2010 to 2020, or like the 2020s, I would say, not just 2025, is this thing called neural networks actually work. And I would say the forefathers like Lacuna or Hinton, Benji, they did a lot of work to establish the foundations. But one guy single-handedly, you know, of course, with a group of amazing engineers who worked with them, truly made it work.

I'd say it's Ilya Sutskever. And I think the magic sauce was throw a lot of data and compute at it. And now you can ask like, oh, wait, is that really all?

Like, was it really that simple? And yes, like, honestly, yes. And I think that's where it came down to blind faith in doing things.

[Speaker 2] (35:53 - 36:51)

I'm sorry, I'm interrupting you again. But can you explain what a neural network actually is? I have a little bit of history with this, because I work in the stock investor world.

And we've had neural networks for a long time. And I remember seeing this over much of the last decade. You would put in a lot of maybe a bunch of different data factors that we have, like maybe time, price, volume, and put all this data into a neural network, try to get it to predict what will happen next, and start maybe a robo advisory kind of service or, you know, try to figure out how a computer might be able to predict.

But none of this played out in the manner that we perceived it when it came to the stock market. But maybe you can define for me, it didn't play out then, I'm talking over the last decade. Can you define what is a neural network in very simple words?

[Speaker 1] (36:52 - 37:37)

So a neural network is a network of artificial neurons connected to each other layer by layer. And what is an artificial neuron is just like a computational unit that takes an input number and gives

you an output number. And so it's called a neural network because it's inspired from the biological neural network, which is the human brain.

But it's not exactly meant to be working the same way either. In fact, that's actually why in practice it works, because a lot of people tried to make it work the exact same way and failed at it. But think about it as like a massive circuit that you're feeding numbers to and it spits out new numbers.

[Speaker 2] (37:39 - 37:45)

Does it spit out new numbers based on the numbers that I have put in in the patterns it recognizes in those?

[Speaker 1] (37:45 - 37:46)

Correct. Yeah, exactly.

[Speaker 2] (37:47 - 38:28)

In the stock market example, if we were to just stick to it, when you put so much data into a neural network and it predicts what might happen tomorrow based on what has happened yesterday, stock markets often tend to be random. And there is a school of thought, they call it technical analysis, where people believe that patterns exist and they try and map out what patterns happened in the past, and how they will repeat themselves specifically. But what if this is a bit selfish, because I'm sticking to the stock example, but what if the past patterns do not recur in the future, then what does the neural network predict?

[Speaker 1] (38:30 - 38:55)

That's a good question. So neural networks can be trained to predict anything, right? Standalone, without the prediction function, the loss function, just the neural network alone is simply a mathematical function, very nonlinear.

Think about it as like some extremely high order polynomial function, right?

[Speaker 2] (38:57 - 38:59)

What was the last word you said? Auto?

[Speaker 1] (38:59 - 39:12)

Extremely high order polynomial function. Right. By that all I mean is like very nonlinear.

A lot of higher order interactions and multiplications.

[Speaker 2] (39:12 - 39:18)

Can you help me picture a neural network? You said it was meant to brain chemistry, but it doesn't.

[Speaker 1] (39:19 - 39:28)

Yeah. Think about it as like, okay, let's say you're feeding in like three or four numbers at the input layer. The first layer will take that and like transform that.

[Speaker 2] (39:29 - 39:38)

Imagine it applies some sinusoids or like- Do you mean when you say transform, are you talking about transformers in Google and their development and stuff like that?

[Speaker 1] (39:38 - 40:59)

No, I don't mean specifically a transformer, but I just mean like a mathematical function, some function F of those four numbers where that function is being learned. But the way in practice it's implemented is there is like a matrix. Those matrix start with a bunch of random numbers in the beginning, and then they multiply with the input you feed.

Then there's some sinusoid or some kind of nonlinear function that takes that and modifies it. Now, why do you need that is because that's where you bring in the higher order dependencies. You're learning that.

And then imagine doing this over like four or five different layers. And then you have a bunch of outputs. It could be four outputs.

It could be 40 outputs. That depends on the way you constructed the neural net. And then there's like a target output that you have based on the dataset.

And the current prediction is taken and the target output is taken. The difference is calculated. And you're updating the parameters of the neural net, which are those matrices at each layer, to update themselves so that you minimize the loss.

But not the loss on one single input, a loss on like a giant dataset, like millions of examples.

[Speaker 2] (40:59 - 41:22)

I can again go back to the stock market example. And when we would put data into a neural network and we didn't get the output we desired, we would go back and curve fit the data in a manner to get a more desirable output. But does it still kind of rely on the premise of recognizing patterns historically?

[Speaker 1] (41:22 - 43:29)

Implicitly, yeah. So if it has to do its job of predicting the output reliably, then it has to recognize whatever patterns it needs to be able to do that. Right?

Like let's say, I'm going to the example of just predicting the next word. If a neural network has to be good at predicting the next word given the previous word, then it implicitly has to understand grammar, sentence construction, common sense, all that stuff. Or if a neural network has to predict the next character in a program that you're writing, it has to somewhat understand the logic, all that stuff.

So it really depends. Whether a neural net captures the useful patterns really depends on what is the task that you're training it on. If you're training it on the raw stock price, let's say you just have a bunch of numbers of the stock price of NVIDIA opening price every single day.

Sure, it's not going to be useful on its own because there are so many other factors that influence the price. And if all it had is like each day's opening price, it's not going to really, there's not really that many patterns in that anyway. So there's this thing in machine learning called the model can only learn whatever actual patterns exist and everything else that's in the data that's noise is irreducible noise.

By that I mean no loss function can hope to capture any of it. You can exactly fit it, but it's not going to generalize. So as long as there's something that's truly signal in your data and the way you crafted the task can capture the signal, like doing the task where it requires you to capture the signal, then yes, the model will definitely be able to capture interesting patterns.

[Speaker 2] (43:30 - 43:36)

And when you said machine learning, I'm sorry again, can you distinguish what is neural network and what is machine learning, the difference?

[Speaker 1] (43:37 - 43:40)

Yeah, so neural networks is one way to do machine learning.

[Speaker 2] (43:41 - 43:43)

How would you define machine learning?

[Speaker 1] (43:44 - 45:22)

Machine learning is broadly train a computer program to do something intelligent or make intelligent predictions on data sets that you're given, such that you're given a recorded bunch of inputs and you want to be able to make intelligent predictions on new inputs that you have not seen before. And the predictions could be anything. And neural networks happens to be a particular way of doing machine learning, where the predictions are done through this abstraction

called neural net that takes in an input and applies matrices, nonlinearities, and then stacks them repeatedly, and then makes predictions out and updates the predictions using backpropagation, the way to change the weights depending on your loss.

So there are so many other ways to do machine learning. There are like support vector machines, linear regression, logistic regression, there's like a whole bunch of techniques. But it happens to be that neural networks is the one way to do things when you really want to benefit from scale.

Like the prediction should keep improving the more data you throw at the problem or more compute you throw at the problem. Neural networks happens to be the most scalable way to do things. But if you have like only 100 or 200 examples, other algorithms might be working as well too.

[Speaker 2] (45:23 - 45:28)

So where does a large language model sit amidst all this? What is it?

[Speaker 1] (45:29 - 45:56)

So a large language model is essentially a giant neural network that's trained on this one task of predicting the next word from the previous word, except it's training on the whole internet. So it's training on terabytes of text, trillions of tokens. And it's doing, it's training on books, code, and textbooks, and like general webpages, news articles, all these things.

[Speaker 2] (45:58 - 46:04)

But the distinction being just text? It's not training on videos and pictures and stuff like that?

[Speaker 1] (46:04 - 46:12)

I think like it can, but since you're calling it large language model, I'm keeping the most people.

[Speaker 2] (46:12 - 46:14)

Let's take ChatGPD for example.

[Speaker 1] (46:15 - 47:11)

Yeah. So the image part, like taking in an image and captioning it and all that stuff comes only in a different phase of training called the post training. But most of the compute is thrown at just predicting the next word from the previous word.

That's called the pre-training. And so essentially think about the dataset. It's the same thing.

Think about the dataset as being the whole internet dump, like all of Wikipedia, all of Reddit,

everything like that. You download it from the web, you tokenize it, that is you convert every sentence into a bunch of tokens, and then you store it somewhere in your S3 dump, and then feed like 4,000 words and ask it to, for each of those 4,000 words, you ask it to predict the next word given the previous word, right? This is where the transformer comes in.

[Speaker 2] (47:11 - 47:11)

Correct.

[Speaker 1] (47:11 - 48:31)

Exactly. This is where the transformer comes in, which is a particular neural network architecture that's pretty efficient. And you shard the model, which is a neural network model on like thousands of GPUs and learn on like trillions of tokens, train this model for three or four months.

And it's pretty amazing artifact emerges out, which is, it'll be great at predicting the next word, but it's still not conditioned enough to be practically useful. And so that's where the post-training process comes in, where you train this or fine-tune this model to be a good chatbot, which is training it to produce good responses to human inputs. And that requires a separate data collection phase where you're collecting data for practically useful tasks, like software programming, compressing emails, summarizing documents, uploading PDFs, and like having it summarize things or answer questions about it.

And then also like just generic conversational outputs where you're training the model to be like conversationally good, keep references of the past and stuff. And once you do that, like you end up with the system like ChatGPT. Right.

[Speaker 2] (48:31 - 48:53)

When I was speaking to Jan, I mean, I asked him things like, you know, explain tokenized to me a dozen times and all of that, but he seemed to think that the current path of evolution of where large language models are going is not the path to AGI. He had a counter opinion on it. Can you elaborate on that a bit?

[Speaker 1] (48:54 - 50:23)

Well, again, like he has his opinions and I think, you know, he's generally been right, so it's worth listening to him. I would say that what Jan wants is like physical common sense too. Like he counts that as a prerequisite for something to be deemed as AGI.

By that I mean like just basic stuff that we all take for granted that we do on a daily basis, which is how to pour water onto a cup, how to like, let's say you're a waiter in a restaurant and you have to pick up like three glasses and two coffee cups with two hands. How do you do it? Right.

Like you're pretty clever. You tilt them in a way and make sure they don't break and so on. Or like, oh, you have a new bottle of wine.

How do you even use an opener that you've never used before? You figure all these things out pretty quickly. The tool used that comes on a daily basis.

You know, like it's not good to mix two ingredients that are not supposed to be mixed. I think these are the things that like he thinks a generally intelligent AI should do, like stuff a cat figures out to just get from one place to another when they're like a maze or like how a rat behaves in a maze to get to where the end state is, despite all the blocks. I think these are things that, you know, like some model like GPT-4 or five can cannot really do right now.

Right.

[Speaker 2] (50:23 - 50:57)

What is the path to that? To that? Like if I think the example of picking up a glass is great because if for me picking up a glass is this easy and if I were to train a computer model to do it, it would require so much energy compute.

And it seems like, you know, you'll have to probably build an arm and figure out how the fingers move and all of that. So the job of a waiter picking up a glass in a restaurant is likely not going to be taken over by a computer.

[Speaker 1] (50:58 - 53:36)

Not anytime soon, which is funny because that's not paid as much as someone gets paid to write code today. Right. So it's like it happens in the reverse way.

Like everybody wants to think what they do is the one that's the one that's going to be taken by an AI the last. So but let's go back to your question. So a lot of things happen in the human brain in a split second.

That's pretty amazing. So the way computers work right now is they would have to watch YouTube videos of people picking up cups and then they would have to have like a physics simulation environment where they train a robot with a suction grouper or like maybe a four or five-fingered dexterous hand attempt to pick the cup like thousands of times or tens of thousands of times and then like learn what its success and failure based on whether the cup was actually picked up or not. And then do this in several different gravity environments so that it generalizes to new settings. Do this on several different visual settings and still it might fail.

If there's a new material that's dropped like a new glass it still might fail. So I think that's where generalization across different physics settings is still like pretty bad. It's not like training on the internet.

There's not enough data. So you actually have to build something that's truly intelligent so that it can learn with very little data. I think we humans, whether you might say it's pretty efficient, but we have had the luxury of evolution.

We've evolved all the basic physical skills that we have like walking, running, doing things with our hands. It's something we've evolved to do over several years. And I think AIs definitely need to spend the compute power to train.

And so we shouldn't compare their training compute to inference compute. And so the best way to solve this problem is like reasoning. Physical common sense and reasoning is what you need.

So you parse the scene and then your planner or reasoner, just like how you're watching like these AI agents now like construct a plan to solve a hard task. I think you ought to do the same thing for physical tasks. Okay.

If I want to pick up three cups together, like this was likely to happen if I do this, that, and then, okay, then this looks like this is the optimal way to do it. And I do it.

[Speaker 2] (53:36 - 53:42)

I think that's also be a similar neural network, which would learn from, I don't know, videos or.

[Speaker 1] (53:42 - 53:56)

Definitely it has to be. It has to learn from videos like, but it also has to build a mental models such that even for scenarios, it's not watched on a video before it should be able to reason and do things.

[Speaker 2] (53:57 - 54:09)

Right. So, Arvind, if I were to ask you what changed, like what changed in the last couple of years that this has taken over everything, like this conversation.

[Speaker 1] (54:11 - 54:46)

I would say it's a lot of compute thrown at the problem, unprecedented scale. The key realization that it's not just compute also, it's high quality data and RLHF learning from human feedback and actually like training it on tasks useful to human labor, like coding and like summarization and stuff like that all came together simultaneously.

[Speaker 2] (54:47 - 54:54)

Do you think the one main thing is throwing immense amounts of compute at the problem without?

[Speaker 1] (54:54 - 54:59)

Definitely that is, if there's like a highest order bit, I think it's.

[Speaker 2] (54:59 - 55:03)

Without worrying if the outcome is going to make up for the revenue spent towards the compute.

[Speaker 1] (55:03 - 55:05)

Yeah. Yeah. I think so.

[Speaker 2] (55:06 - 55:07)

Is that the distinction?

[Speaker 1] (55:08 - 56:53)

Definitely. Because like the compute and by the way, compute alone is useless. Like people have tried to reproduce these things by doing the same thing and it doesn't work.

You got to throw high quality data tokens at the problem too. So that taste on like curating datasets of like what will really matter. Like for example, if you want reasoning to emerge in a model, it's good for you to like make sure you have YouTube transcripts of video, like lectures, MIT lectures, Stanford lectures, and textbooks like where you actually have problems.

It's not just a problem, but the solution is explained step-by-step. So when the models learn this, you can actually prompt engineer them at inference time to say, think step-by-step and then it's able to think step-by-step and solve a problem. Then that leads to the next idea, which is chain of thought, where you train the models, you collect the dataset with like chain of thought, where the model is not just solving the problem, but it's actually understanding why something is right and wrong.

So even if it's wrong, it can try to like go back and rethink and like iterate and improve itself. So I think like there are like a bunch of three or four key ideas that came one step at a time, and they all stacked on top of each other. But the main realization is high quality datasets with a lot of compute and trained to be conversational with human feedback, made accessible to all people through a simple chatbot interface, made magic happen.

I think I think it's a lot of like four or five good things coming together.

[Speaker 2] (56:56 - 57:06)

Great. I think I have a better understanding of where we are at right now. I want to digress for a minute.

You said Bangalore. How long were you there for and what were you doing? That's where I'm from.

[Speaker 1] (57:08 - 57:08)

You mean for my internship?

[Speaker 2] (57:09 - 57:12)

Yeah. You said you solved the problem in three weeks.

[Speaker 1] (57:13 - 57:32)

Yeah, probably. I think I was in this place called Koramangala. And I didn't actually explore.

So I just worked all the time, which is, you know, now that I look back, I probably think I should have explored.

[Speaker 2] (57:33 - 57:39)

But now when I look at you, I think you did a good thing by not exploring Koramangala and working all the time.

[Speaker 1] (57:40 - 58:10)

No, not not not. I don't mean to explore Koramangala, but I just mean explore Bangalore in general that I wish I did. But I do remember the traffic being bad and I'm being told it's even worse now.

So so probably not probably good that I stayed in the room and just worked. But otherwise, like I do remember the weather was awesome compared to Chennai. I think weather was much better.

What did I do?

[Speaker 2] (58:10 - 58:10)

Do you still follow cricket?

[Speaker 1] (58:10 - 58:15)

Uh, yeah, I do. Yeah, I followed the match on Sunday.

[Speaker 2] (58:15 - 58:20)

Yeah, I'm actually in Dubai because I came to watch the match. There's a hotel room in Dubai.

[Speaker 1] (58:20 - 58:21)

Oh, cool.

[Speaker 2] (58:21 - 58:41)

It was good. It was good. I feel like the stadium had maybe 99.99% Indian support and 0.01% New Zealand. They had like a tiny box with 30 people in there. Wow. So everybody walked away happy.

Most people walked away happy.

[Speaker 1] (58:41 - 59:03)

Yeah. Yeah. I mean, like I I was pretty disappointed in the last three or four times we lost in the semifinal or the final.

So I was really hoping that India wins this time. So that was awesome. But honestly, like I want I want India to win the 2027 World Cup.

I think that'll be pretty big.

[Speaker 2] (59:04 - 59:07)

And you were saying about Bangalore. What do you remember of it?

[Speaker 1] (59:10 - 59:14)

To be very blunt, I only remember that I worked really I worked all the time.

[Speaker 2] (59:15 - 59:16)

Have you been like this all your life?

[Speaker 1] (59:18 - 59:24)

I mean, like I worked, I worked pretty hard. I'm very proud of that.

[Speaker 2] (59:24 - 59:29)

And is there a why you work very hard and you're proud of it?

[Speaker 1] (59:29 - 1:00:07)

I think I enjoy it. I'm not doing it because, oh, like you got to do this and then you'll achieve that. And because that's that's impossible to scale.

That's how most people are when they when they're studying for IIT or like when they're trying to study for grades to get good grades in IIT. Most people are that way where they do it because it'll

get them a reward. I think some of it applied to me, too, but I mostly do it because I enjoy it.

And I think that's why I'm able to still keep doing it.

[Speaker 2] (1:00:08 - 1:00:21)

Which part do you enjoy? Is it the is it the stuff? There's so many things you've gotten at this end of the bridge.

Which part have you enjoyed the most and which part do you enjoy most today?

[Speaker 1] (1:00:21 - 1:00:27)

I think I enjoy the intellectual part of like learning new things, being curious and learning new things.

[Speaker 2] (1:00:28 - 1:00:30)

You're going to be disappointed in this conversation.

[Speaker 1] (1:00:32 - 1:01:56)

Well, I you know, one thing they say is like. You might think, you know, something until someone asks you the most basic questions and then you're like, OK, let me let me figure out a way to explain it. Let me figure out and then you're truly testing the limits of your understanding.

I actually enjoy these kind of conversations. I had a similar chat with Lex Friedman. He made me go through the whole like history of like AI and neural nets and like search and you know how like Google makes money.

And I'm like, wait, I actually thought I knew all this, but this guy is really testing and making me question if I really know stuff. And I enjoy those kind of conversations, actually, because it's pretty rare to talk to. And it's not like you make a list of the easiest questions, right?

You're actually trying to go deeper and deeper. And this is also why we even built this product. It's letting people do that on their own and no questions.

There's even a saying from Confucius, right, that you might feel like a fool if you ask a question, you know, supposedly a simple question, you might feel like a fool for a minute, but you'll be a fool for your lifetime if you don't ask it. And so I think I'm actually like always in favor of people asking questions.

[Speaker 2] (1:01:57 - 1:02:03)

I'm going to use that for the rest of today and ask you more stupid questions. Is it going?

[Speaker 1] (1:02:04 - 1:02:41)

Yeah, hopefully I can come out answering them well. But yeah, I genuinely enjoy like learning things. I enjoy the challenge of trying to do hard things.

Like in general, my whole life has been trying to do something that seemed like pretty impossible. I'm not necessarily from a rich background. So most of the stuff we did, you know, myself, my parents, to get here, get into IIT or like get into Berkeley, you know, get a job at OpenAI, like start this company.

[Speaker 2] (1:02:41 - 1:02:42)

Can I ask a question?

[Speaker 1] (1:02:42 - 1:02:42)

Yeah.

[Speaker 2] (1:02:43 - 1:02:52)

When you spoke just now, you said we did, got into IIT and got here now, we got here now. Do you think of your family, your parents that way?

[Speaker 1] (1:02:53 - 1:04:30)

Yeah. I mean, like I did all the work to study and, you know, do the exams well, but they took care of the other stuff for me. Right.

And so it's not a individual thing. Same thing now, like I'm doing the work running the company, but my wife takes care of so many things for me at home. And it's not just about the support at home or something.

It's more the moral support. I think you have very few people to lean on to. And so there are so many times when you're not like necessarily feeling the best about your chances.

And there's a lot of things that you cannot share with your own fellow colleagues, because, you know, as founder, CEO, you always have to appear like I have it all figured out. So there's somebody you need to go and talk to for help. Or someone to push you also, right?

Sometimes when things are going well, you might feel like you're on the top of the world and someone has to bring you back to earth and say, hey, calm down. You have nothing figured out yet. Who does that for you?

My wife does that for me. And then back when I was studying for IIT, my mom was always keeping me in check and making sure that I was focused. And it's important.

You can't have too many people doing that for you at the same time. The more people do that, it gets chaotic. So it's good to have one or two people doing this all the time.

[Speaker 2] (1:04:30 - 1:04:32)

What did your parents do, Arvind?

[Speaker 1] (1:04:33 - 1:05:11)

My mom works in the government, central government. My dad was an accountant. So we financial accountant.

So we, I'm actually the first engineer in like extended family. And so Really? You're Tamilian?

Yeah. Correct. Yeah.

But our family had more the accounting background. And so engineering was still like a new thing at the time for us.

[Speaker 2] (1:05:12 - 1:05:54)

Our audience in this particular thing, we speak to wannabe entrepreneurs from India, largely who are under the age of 25. But for a second, I'm going to put on my investor cap and ask you what the big players are doing? How do you distinguish one from another?

And maybe you can give us a bit of nuance of how is one different from another? Like you take a grog, you take what a Meta is doing? Yeah.

You talk about what Microsoft is doing? Yeah. Maybe like, just like, you know, like really low level stuff that I can understand?

Yeah.

[Speaker 1] (1:05:54 - 1:07:47)

Honest answer right now is all of them are doing similar things. Okay. Like, I'll just say it as blunt as it can be, is there's not really a genuine differentiation between ChatGPT or Anthropic or Gemini or Grok or Meta AI right now.

And of course, for Perplexity, you can argue similarly, which is, you know, in the beginning, the differentiation was we were the only ones to make sure you always had sources for everything and like, you know, highly accurate sources, fast answers and so on. But everybody else is also like realizing that the real values in search, even more than free form chat, and they're trying to put sources for almost any response. So I think right now we're in this weird phase where like all AI chatbots seem similar and like some people prefer one over the other.

And like, you know, if you rank response accuracy, like I'm sure different benchmarks will have different ranking one number one or two. But consistently, Perplexity is deemed as like one of the most accurate, fastest chatbots. And I'm very happy because that's the work we've put in the last two years.

But I feel this year in 2025 and 6, the differentiation is going to come from more agentic behavior, where like the question answering, like answering questions will be seen as a commodity. Some people have preferences for some products, some user interfaces are going to be better. Those will not just respond with text, but also give you charts and like images and inline product cards or hotels.

[Speaker 2] (1:07:48 - 1:08:00)

Would you call that agentic? Like if somebody picks off a language answer, like a text answer from a certain large language model and converts it into images and makes it more...

[Speaker 1] (1:08:00 - 1:09:12)

No, that's not agentic. I'm just saying question answering itself, you can say just responding in text is like not going to cut it. Like let's say I'm going to ask for the best shoes, you want to actually see the shoes.

You want to see an actual shoe card and like reviews and like compactly summarize to you like options to buy. Same thing with hotels, same thing with like restaurants. You can just book it right there.

I think these kinds of experiences will differentiate one or two chatbots from the rest and we are doing our part to be ahead of the curve there. But I feel like the real magic is going to come from AI doing things, where you can go to the AI and ask it to play a song or like play a video or book a restaurant reservation, book an Uber, book a flight, send an email, move your calendar. Like say I'm communicating to Nikhil's team, I'm just going to ask my assistant to like, hey, can you ask them, can we start at 8.30 instead of 8?

And it's just going to do this emailing for me. And it's going to do the back and forth with your team and it's just going to like figure it out. I'm just like in my bed sleeping and the AI is working for me.

I think those kinds of things...

[Speaker 2] (1:09:12 - 1:09:16)

Why is that piece missing? Why is it not happening now? What needs to change?

[Speaker 1] (1:09:17 - 1:09:35)

It's only recently began to take off because reasoning only recently began to work. And without reasoning, you cannot do these things. With just the LLM in the traditional sense, where you get an output for an input, it's very hard to do these things.

[Speaker 2] (1:09:36 - 1:09:43)

Reasoning based on my data or reasoning based on generic data with nuance derived from Yeah.

[Speaker 1] (1:09:43 - 1:10:43)

So context comes from you. But the core reasoning skill is in the model. The context of like, okay, your emails, your existing calendar, we need access to all that.

And it needs to be contextual. And I think that's why product building is equally important or probably even more than the model here, because there's going to be a bunch of great reasoning models, but there's not going to be 100 products that really package personal context, all the API integrations, services integrations, native integration to your phone to be an assistant really well, voice experience. Earlier, there'll be like 10 details to get right.

Now, there'll be like 50 to 100 details to get right. And the more details to get right simultaneously, the less chance that there are like five or six different chatbots doing the same thing.

[Speaker 2] (1:10:44 - 1:10:54)

Eventually, if all data is democratized, and all the models consume all the data, will everyone throw out the same answer, but the language be different?

[Speaker 1] (1:10:55 - 1:11:26)

It's already that way right now. If you leave the search part and just talk to an AI and ask questions, most of these models are kind of saying the same thing. And there's one reason for that is because they're all trying to climb on the same benchmarks, the same leaderboards.

So there's not a very big qualitative difference. You can squint at it and say, okay, I like the response style of this one model or the other, but it doesn't matter.

[Speaker 2] (1:11:26 - 1:11:48)

So what is the nuance that makes one subscribe? I'm a user of Perplexity, I have Perplexity Pro, I have ChatGPT, I have a bunch of other models. So what nuance will attract me towards Perplexity versus ChatGPT versus Grok versus Metas AI?

What is the difference?

[Speaker 1] (1:11:49 - 1:12:42)

I think it really depends on what you're using AI for. If you're a person who uses AI for like a lot of fact checks and research and sources and financial research even, you want to get charts, you want to get stock prices, balance sheets, all that stuff. Whoever does this best, you will subscribe to it, right?

Who does it best now? I think it's us, but if you don't think so, I would love to know. But at the same time, here's the thing.

I feel like our product has this advantage that we can use any model out there. It's kind of weird, you could ask why can't ChatGPT also have Grok in it? I think it's more that they also have a different rivalry going on of who trains the smartest model, which is what attracts the researchers to be working there.

[Speaker 2] (1:12:44 - 1:12:47)

How do you pick which model you use for which answer?

[Speaker 1] (1:12:47 - 1:12:51)

We regularly evaluate models on so many different types of queries.

[Speaker 2] (1:12:51 - 1:12:57)

You're using one model at a time and then you switch to another? It's not that one query goes to a certain model?

[Speaker 1] (1:12:58 - 1:13:29)

It's neither. Every query goes to a bunch of models, but they're doing different tasks. One model rewrites your query into a more easily understandable format for the AIs.

Another model does the chunking of the pages into parts that gets consumed by the summarization model. And the summarization of chat models is different. And then there's another model that suggests new questions to ask.

So these are all four or five different models working per query.

[Speaker 2] (1:13:30 - 1:13:37)

Is there a latency play in that? If you're running on top of so many models, will you be relatively slower?

[Speaker 1] (1:13:38 - 1:14:07)

No. That's why even though people say it's a wrapper, there's a lot of back-end infrastructure work we put in to make it so fast. I still think we are the fastest product among all these because the latency is one of our main metrics we track internally.

Tail latency, actually, there's a concept called tail latency, which is a 99th percentile that matters. It's not the mean latency.

[Speaker 2] (1:14:08 - 1:14:18)

Is that if you throw a part of the answer out and then there are sub-questions which you click and then you go to the next parts? Unlike another model which throws everything out at the same time.

[Speaker 1] (1:14:19 - 1:15:41)

So the trick that most people figured out in AI chatbots is you stream the answer, a few chunks of words. That way, the user doesn't feel the latency. They're just already beginning to read the answer.

It's a clever hack. This happens in voice-to-voice also, by the way. The reason it feels real-time is the answer is still not done yet, but it's begun talking to you already and you're just hearing it.

One thing that we try to do is a lot of the open-source models that we serve ourselves with some fine-tuning, we've tried to serve it with extreme efficiency. We wrote our own runtimes for NVIDIA chips and we use other chips like Cerebrus. That helps us to make the latency as low as possible.

The fact that we have our own index lets us pull the links really fast with sub-second latency. The overall latency feels really short, even though we do a lot more work on the backend. But I think there's still some more juice to squeeze out here.

I feel like in the end of the year, there'll be another half a second that's shaved off here with more improvements on the infrastructure.

[Speaker 2] (1:15:41 - 1:16:00)

To be really honest, as a user, I don't even care. If one answer is coming at 300 microseconds and another is coming at 800, and if my naked eye can't tell the difference and you're streaming anyway, so I'm reading, I don't know how many people truly care about that difference.

[Speaker 1] (1:16:00 - 1:16:53)

That's the thing. You might feel that way. We don't get 10,000 requests, but let's say we get 100,000 requests per second in future.

I think these things will matter. Because of the load, you will start to feel it slower too, even though right now you don't. That's where any day, Google has done this historically.

Anytime they shaved off even 100 milliseconds on the Google search result page load, like the loading time, they've always measured that retention increased. We don't care. It doesn't matter.

But then at the scale of traffic you serve, the more improvements you do, the better. Because that way you can handle any tail cases pretty well.

[Speaker 2] (1:16:53 - 1:17:15)

Makes sense, actually. Is there a way to, I know this is probably a very hard thing to extrapolate, but is there a cost per query to service that you have versus a fee that the user pays? Like if I pay 20 bucks a month to Perplexity, how many requests do I need to make?

[Speaker 1] (1:17:15 - 1:19:27)

To really consume that? I would just say that it's not something that's a static metric anymore, considering every three months or something, there's a new open source model out there, and then that forces the model apps to lower the prices because then nobody's going to use their APIs. So it's something that's constantly going down, the cost.

That doesn't mean that we get to mint more money from you either, because what happens is there are newer experiences that are being built with the more expensive ones simultaneously. Like the deep research stuff, it's actually pretty expensive to serve deep research for us. We still price it at $20 a month.

I think OpenAI's one is slightly more detailed on some queries, but it's priced at $200 a month. And you can see that that's simply because DeepSeq is open source and we're able to actually serve it at 10x cheaper price. And we'll address the inaccuracies in the next few months, like with better models, more fine tuning.

So I think that actually makes the margins lower for us on the proscription, if people use deep research. But while this happens, the cost per query on regular pro searches or reasoning searches go down because there's more progress on the model side. And I think that's agentic tasks, like when these AIs start beginning to do stuff for you, it'll definitely cost us more.

And so we're actually okay with this uncertainty in what the real margins are on consumer subscriptions in AI in the short term, because I think the real thing to focus on is getting the experience really good. And sure, we would love to not burn the money all the way, but at the same time, hyper-optimizing for margins now would be the wrong tactical move.

[Speaker 2] (1:19:28 - 1:19:45)

If I were to hold you to a specific answer, if you had a hundred bucks and you could only put it in one company, you can take out the private ones, so don't put yourself in the list. But amongst the listed players, if you could invest a hundred bucks in them, which one would you pick?

[Speaker 1] (1:19:49 - 1:19:51)

In an AI or any company?

[Speaker 2] (1:19:51 - 1:19:56)

Yeah, AI, because these guys do everything now.

[Speaker 1] (1:19:56 - 1:21:56)

I think I would put it into meta, mainly because in a world where AI works increasingly well, like, I think the human to human connection becomes even more essential. And there's literally no one disrupting that in Instagram or WhatsApp. And so advertisements, and in a world where people are going to be able to ask AIs to do stuff for them, brand value, how much a brand themselves get known to the user matters even more.

Because you can ask the agents to just ignore all the sponsored links on Google and truly look for what's best, read the reviews and stuff. So I think what people perceive a brand as matters even more, and people-to-people connection matters even more, and people knowing what other people say matters more. So I feel like they're very well positioned to keep their existing ad business strong or make it even stronger in a world where AIs actually work.

It's a kind of an interesting position to be in for them where their ads business is going to flourish even better when AIs work. I wouldn't say the same for Google. I think Google Ads and Google Agents and Google Ads are just completely on the opposite ends of business incentives.

And that's kind of why also Google has the least incentive to bring out AI native search or agents right there on core Google home page or Google Apps. It can be hidden in a mode or sometimes firing for some queries, but it's never going to be the central piece of it. Meta doesn't have this problem at all.

They can roll out AIs, but the core feed of watching what other people post is still going to be the same.

[Speaker 2] (1:21:57 - 1:23:33)

I was thinking about this the other day, with all the talk about tariffs and who imports how much and who exports so much and the deficit that the US is running versus India or China or many of these other countries. Almost every new company here in India spends all of their marketing money, their distribution money. Like if I were to even start a t-shirt company, a coffee brand, I

don't know, a SaaS company.

Gone are the days of putting an ad on a newspaper or a TV channel or a cricketing team or you know, like the traditional way of spending money to get distribution. Don't hold me to the numbers, but pragmatically I see that declining and more and more money in discovering clients in India goes to either Facebook or Google and Meta or Google and that money, that revenue might be registered in Facebook India or Meta India or Google India. But essentially, the way I look at it, it's trickling back to the parent company because the market cap of these companies are going up by virtue of the revenue they register here in India.

Since we are at the very core of it, talking to entrepreneurs who want to start something in India, do you think there's a play there to disrupt this market? Do you think that's even remotely possible for an Indian, for someone in the Indian youth to build something to take away some of this pie?

[Speaker 1] (1:23:35 - 1:24:41)

Well, if an Indian company started an Instagram or WhatsApp rival, I would be very impressed by the bravery of it. Not that like I'm trying to do anything smaller. No.

Well, I would say like what I'm doing is similar, where even now people think it's a stupid idea to compete with Google. But I think there's some angle that can work. And here's how I would see it.

If you can build way better targeting than Instagram does, at least for consumers in India that you're trying to target for your business. And sure, like people would be at least, all they got to do is like if they're spending a million dollars in ads a year, instead of spending the entire million on Instagram, if they spend 700K on Instagram and 300K on your thing, like that's already a big disruption.

[Speaker 2] (1:24:41 - 1:24:46)

That's step two after I first garner the distribution for my platform where people come from.

[Speaker 1] (1:24:47 - 1:24:53)

Yeah. And you need to have one core reason why people even post on your platform.

[Speaker 2] (1:24:54 - 1:24:54)

Yeah.

[Speaker 1] (1:24:55 - 1:25:52)

And when you have zero users or you're just like getting users, the creators are like, you know, the ones who are posting stuff, they want traction, they want likes and stuff, they want shares.

And so that's the problem. That's a cold start problem.

And network effects, that's kind of why I said Meta has a bigger moat than Google, because Google's moat on distribution comes from their deals with carriers, OEMs and all these people. But Meta's moat is this raw network effects. Nobody pre-installs Instagram or WhatsApp on phones.

Android pre-installs Google. Despite that, everybody goes and installs these apps. So I think you got to change that in some way.

And you got to build a user base from scratch. And I think that's the hard part.

[Speaker 2] (1:25:53 - 1:25:58)

Any ideas? See, I want to try it. How do I do it?

Give me an angle.

[Speaker 1] (1:25:59 - 1:27:31)

To be honest, I haven't thought hard about it. But let me just try to think on the feet here. The last big app that actually did it was TikTok.

And interestingly, they actually grew a lot through Instagram. They spent billions of dollars of ad spend on Instagram to grow TikTok. And what I was told is the Meta team was laughing at it.

Like, hey, we grew all these users organically. And little did they realize that the retention on acquiring users through paid channels is pretty low. And they're just making us rich and growing our stock price through increasing our ad revenue.

And they're not going to actually retain any of the users. It doesn't matter. But that ended up being wrong.

They actually got a lot of users. And the only reason Instagram is still fine is because TikTok is banned in many countries, particularly in India. So I would say, definitely, you got to spend a lot per user on existing channels.

And definitely, you got to have some new unit of information that existing platforms don't have that becomes core to your, like Reels was new. Obviously, Instagram copied it. But at least for a while, it was new on TikTok.

[Speaker 2] (1:27:32 - 1:27:36)

Is that where perplexity will go eventually? Do you think you'll need to have ads?

[Speaker 1] (1:27:37 - 1:29:11)

I hope not. I think the market for like, just like an assistant that is so personalized to you, and does a lot of work for you, gives you daily briefs, updates, does market research for you without you even asking for it, is massive. Like, people would pay like hundreds of dollars a month for such an assistant because it's kind of like hiring a person.

And if the hour poop per person is so high, like $1,000 a year, and if we can at least get 10 million people to pay for it, I think that's like a pretty successful multi hundred billion dollar company of its own. And if you can do that and figuring out a way to like grow X percent a year and getting to Google's like ad revenue, our magnitude is certainly more achievable. And so that's, I also think when an assistant's really personalized, ads are pretty easily doable.

Like, you know, the reason Instagram ads are better than Google ads is they're very personalized to you, right? And so Instagram has done some research that if they remove the ads on the platform, the engagement time went down. Google has never really publicly done any study like that.

I'm actually certain that like, Google removing ads makes the experience better today.

[Speaker 2] (1:29:13 - 1:29:14)

But can they survive without ads?

[Speaker 1] (1:29:14 - 1:29:15)

They cannot. That's the thing.

[Speaker 2] (1:29:16 - 1:29:17)

Yeah, yeah.

[Speaker 1] (1:29:17 - 1:29:18)

So that's where the...

[Speaker 2] (1:29:18 - 1:29:39)

I've kind of like stopped using Google for so many things. You know, what I find today is I will search for something on Google, I will get annoyed by all the ads, and then I immediately open up, you guys are open AI and I search there because I feel like I don't have to like, in Google you're almost searching and then searching again.

[Speaker 1] (1:29:40 - 1:29:46)

You are, you are searching first on Google. Yeah. Why?

Because they are the search bar.

[Speaker 2] (1:29:47 - 1:29:48)

Yeah.

[Speaker 1] (1:29:48 - 1:32:36)

And that's why you got to build a browser. Or you can convince people to go change your default search engine to perplexity. But you know that what they do, they always put these pop-ups that say, hey, turn your default back to Google.

And then in that, there'll be two options. And the highlighted blue one would be yes. And this, like retain back would be the non-highlighted option, they'll be like non-bold text.

So they have all these like tricks they learned through ages to like resolve their dominance, that the first query goes to them. Another thing I'll tell you why they'll still continue to be dominant for a few years is, let's say you do your research on like what microphones to buy, or like, you know, let's say the best headphones for podcast recording, you're buying equipment for your podcast studio. You do your research on perplexity or chat GPT, you know what to buy now.

You go and actually make the purchase on Google. Or you go to Amazon. Like, you know, most people just type the brand on Google, they click on the Amazon link and then they go and buy there.

So who actually makes money out of that research you did? Google. Like not us, right?

Because Google makes some money every time you click on a link and make a purchase. Because they get to claim cost per click conversion to the advertiser. And the advertiser is like, oh, wait, I got, I'm getting all this, you know, conversions because of Google.

So I'm going to keep spending advertisement revenue on it. So this is the problem. Like you got to have AIs that not just help you do research, but help you make transactions natively.

Like, and then you got to have AIs that are not, you know, vulnerable to the search bar placement. And that's the real challenge that, you know, companies like us or chat GPT have to address. It's not the fact that they can't provide a better product.

I think it's pretty obviously clear. I think you got to be able to finish the other two or three steps that remain to get rid of the Google dominance. And Android is a massive advantage for them.

They don't let people sell a phone if you don't keep Google as a default search engine. What they do is like they say you cannot have the Play Store. Okay, if you don't have the Play Store, there are no apps because nobody is building apps on any other Play Store.

And so no phone maker can sell a phone there. So and they don't share ad revenue on the Google search, if you don't have the Google Assistant as a default. So there's a lot of things.

[Speaker 2] (1:32:36 - 1:32:41)

It sounds like a relatively easier business to disrupt the Play Store or the App Store.

[Speaker 1] (1:32:43 - 1:34:10)

First of all, like what most of the people are building apps, you can have a fork of Android and ask people to publish it to your App Store too, but they don't get the visibility that they get on the Google's Android, because the phone makers are still going to use Google's Android. And the phone makers are using Google's Android because Google shares with them the ad revenue on Google search that's getting made on the mobile phones. And you cannot do that until you have the scale.

So it's very tied to like many things. And the more you understand the details, like you're always like playing chess, figuring out like, okay, like what is the next move you can make that you can make as a startup with way lower resources and still convince the telcos and OEMs to work with you. I think that's the hard challenge.

And they're always there to like spoil your plans. But again, like I expect to the question you asked about, can I have an Indian rival to Instagram? You're going to have to play, you're not just going to be focused on building a better product.

You have to actually spend like a lot of your energy thinking about distribution. And Matt can anytime come and say like, hey, like just pre-install Instagram on all your phones, and I'll share ad revenue with you for Indian users.

[Speaker 2] (1:34:11 - 1:34:16)

And even before that, I need an angle as to how I get the initial user base.

[Speaker 1] (1:34:17 - 1:34:40)

Exactly. So it's very difficult, but I certainly think it's worth the, like, you know, of attempting some, some brave person needs to do it. Ideal is the person who begins it as having their own audience so they can actually influence.

Silicon Valley people have tried it, like Clubhouse. Again, everything has lessons you can learn from.

[Speaker 2] (1:34:43 - 1:34:50)

So the question is, if text moved to pictures moved to short form video, could it be long form

video?

[Speaker 1] (1:34:50 - 1:34:52)

No, YouTube, YouTube is there, right?

[Speaker 2] (1:34:53 - 1:34:53)

Yeah.

[Speaker 1] (1:34:53 - 1:35:00)

But the YouTube is actually one of the biggest rivals to Instagram, I would say.

[Speaker 2] (1:35:00 - 1:35:09)

Mm hmm. Mainly because that was Instagram's market to take, right? The real long form video, the long form video as well.

[Speaker 1] (1:35:09 - 1:35:56)

Long form video. Yeah, yeah. A little bit.

Yeah. What I was told is YouTube's ad revenue now comes more from TVs than even like the actual YouTube app. So, so you can kind of see where people are actually beginning to spend more time on the TVs now than even their mobile apps.

So there's probably something there. Podcasts is growing a lot. A lot of people, of course, you're making one of the most listened to podcasts, but it's a, it's a, it's a thing that Instagram is not really getting.

And it's going more towards the Apple podcast, Spotify. So there's always like people figuring out new forms of content that's not necessarily going to Instagram.

[Speaker 2] (1:35:56 - 1:36:20)

So that's something that's an opportunity. If I'm able to aggregate every Indian podcaster and improve the quality of their video by, I don't know, if I include a chat function where they can talk to the podcaster and the guest, like have some angle like that. Do you think if I were to be able to aggregate that, is it a possibility?

[Speaker 1] (1:36:21 - 1:36:49)

Definitely. Another thing that people haven't really tried is like live stream the podcast, like, let's say we're talking now. And like, so the way podcasts work is we record it, you edit it, you post it, and then people are listening to it, but there's no communication between us and them.

Right. And like X tried that with, with live stream. And you know, like, so there's something...

[Speaker 2] (1:36:49 - 1:36:54)

Instagram has it too, Instagram live, but it's not really podcast, right, right, right. Yeah.

[Speaker 1] (1:36:55 - 1:37:24)

But there's something where like, you can consume all the podcasts, you can also talk to the people who did the podcast, they will respond to you on the comments. You can, you can probably like, I'd say, Hey, I want to hear this Nikhil's thing in like, but only the parts where he talks about AI. And it'll just edit it like really fast and just make a new version.

Just, you just listen to that.

[Speaker 2] (1:37:25 - 1:37:30)

I don't think YouTube doesn't do well. What do you call it when you convert video to text?

[Speaker 1] (1:37:31 - 1:37:32)

The transcriptions?

[Speaker 2] (1:37:33 - 1:37:35)

Yeah, they don't do great at transcriptions, right?

[Speaker 1] (1:37:35 - 1:37:59)

Yeah. But again, I'm just saying like, why do you even need to see the transcript? Transcripts are there because it's a hack to get to what you want.

But if you literally just enter a prompt and say, just make a version of this podcast for me that edits out what Arvind and Nikhil talk about AI, or like neural networks, and it just creates that segment. And they just listen to that.

[Speaker 2] (1:37:59 - 1:38:05)

I thought most large language models are text, they're not consuming video yet, right?

[Speaker 1] (1:38:06 - 1:39:11)

Exactly. So you don't need the video part as much, you just need to make sure the transcript is pretty accurate. Or even take the MP3 file, the audio file, and then the long context is good enough to consume all of it.

And then you just say, I want only these parts out and it'll tell you the timestamps and you take that and make a video out of it. It's going to have rough edges, I'm sure it's not going to work perfectly. But with some engineering, you can make something this happen.

The hard part, honestly, Nikhil, is you got to start from scratch, you got to create incentives for people to consume stuff. So there's something new element needed, and then a lot of sharing on existing platforms of how this is the next big thing. But you're right, if there's one way to aggregate all the podcasts that's happening in India on one platform, and people being able to edit it, listening in any new language they want, on the fly, it might be a big product market fit there.

[Speaker 2] (1:39:11 - 1:40:13)

Very interesting. I'm going to ask you another personal question. I have a private equity fund, we're reviewing a data center business, fairly large, something that does maybe $100 million of EBITDA right now.

So data center has become such a thing in India, Arvind, every real estate person that you speak to, I speak to today, everyone's talking about data centers, it's like, it's like the real estate almost in the 2025 version, the big thing for them is not this new building, but it's building a data center. If you're able to buy data center businesses at a 20 multiple of EBITDA or a 25 multiple of EBITDA, would you do it today? Is there something missing?

Is there something changing in terms of how data is being compressed, quantum computing, or compute moving out of the data center, that one should not do it?

[Speaker 1] (1:40:13 - 1:42:28)

I wouldn't really worry about quantum computing right now. I think it's still pretty early stages. I certainly think India should have its own data centers, like there's no reason not to.

And definitely calls for good real estate expertise. Infrastructure build out is not easy. Buying the chips, connecting them, making sure you use the right technology for the interconnects between these different GPUs, building these server racks, I mean, compute centers in different IITs have done this.

We had our compute cluster that we had access to in IIT, and it would live in the computer center. So definitely doable. And it really depends on, okay, so there's this company called CoreWeave in the US, I think it's going to IPO pretty soon.

It's the first pure data center play that I've seen. It's not a big tech data center. NVIDIA owns a big chunk of this company.

And I think the way they compete against the rest is they do the build outs faster. And OpenAI is using them, a bunch of others are using them. So if you can provide training GPUs to people in

India, much faster and cheaper prices, potentially cheaper because the data center build out costs might be lower, because labor costs are lower.

There's probably something there. I hope at least for inference, it makes a lot of sense because data sovereignty might be a thing. So let's say even for companies like Austin Future, if the government of India wants the data of people using Perplex in India to stay in India, then it makes sense to have even American companies or other companies outside India to be using the data centers built out in India so that the data is stored in India.

[Speaker 2] (1:42:28 - 1:42:54)

I think it'll happen eventually, invariably. Now the financial data sits out of India. And India creates something like, I don't know, 20% of all the data because of the number of people with smartphones.

So the assumption is it will happen and hence everybody's talking about the data center business. But structurally, there is nothing that is changing in the data center business.

[Speaker 1] (1:42:55 - 1:43:01)

I don't expect it to be a pretty high margin business of its own, unless you combine it with good software.

[Speaker 2] (1:43:03 - 1:43:05)

And what would software look like for a data center?

[Speaker 1] (1:43:06 - 1:43:34)

Is it like spin up jobs, easily host models, have the Kubernetes support for like scaling instances? That's kind of what the cloud companies have shown, right? Maybe in the short term, if you're the only one who can provide a data center in India, you're going to enjoy good margins.

But long run, you should expect more people playing the game.

[Speaker 2] (1:43:35 - 1:44:05)

Yeah, no, there are many providers already. There is maybe a gigawatt worth of data centers. I mean, I'm not sure of the exact number.

But it has scaled significantly. The question is, does it continue to grow in this manner where at the end of the day, it's a very commoditized business. It's almost like a real estate company starting a warehouse.

I'm not able to distinguish if one has IP over another.

[Speaker 1] (1:44:05 - 1:44:10)

Not really, until you have like some vertical integration done pretty well.

[Speaker 2] (1:44:11 - 1:44:18)

And the other big worry is, does this become such a big business that the hyperscalers build their own and do not go to a third party vendor?

[Speaker 1] (1:44:19 - 1:44:56)

Possible. I mean, hyperscalers actually build their own data centers everywhere, except where there are like real constraints where they have to move super fast. And the only way to do that is to work with someone else locally.

And there are like local regulations and restrictions on what other companies can do in physical spaces. And someone who's already there, who's an Indian business, can be the only one who can do that on the timelines they want.

[Speaker 2] (1:44:58 - 1:45:11)

Do you have a view on Nvidia? Like, the margins that they operate at and the scale of revenues and profitability they are at, why has there been no disruption?

[Speaker 1] (1:45:12 - 1:47:28)

I think it's pretty hard to do what they do at the margins they have. That's the main reason. They have a very flexible chip.

It can do a lot of computations. It's not just about inference. It's not just about training.

It's not just about like dense models or mixture of expert sparse models. It's not specialized, so it's very general. So you can do a lot of things with one chip.

And they have perfected the art of like the interconnects, the data center buildouts. And I think software is a big advantage for them, too. The fact that CUDA has a big moat.

And people, developers, are all like trained and programmed to already learn to use CUDA. And it's very hard to go learn a new software stack. And they keep a lot of the CUDA stuff, like closed source.

It's pretty hard to replicate it. And then by the time you do all the work, going and building your own software stack and your own hardware and making it pretty general, they have the next generation of chips. And then they already have the relationships with all the hyperscalers to get their chips in as first priority.

So they are competing on many levels. It's pretty difficult. But recently what's happened is, at least on the inference layer, there are like some alternatives, like Cerebros is there and Grok with the Q.

And again, they're enjoying the time period before Blackwell comes to the market. And Blackwell chips are supposed to be way more efficient than H100s for inference. So maybe all the things are going to be short-lived.

We never know. Certainly the margins might get affected, but also the raw AI usage and adoption and how many others are going to build AIs is also going to grow. The company might still be a very lucrative business to invest in, but it's one of the least understood stocks, I would say.

Even though there's a lot of energy and effort being put into understanding it, it's one of the least understood stocks. And it's pretty volatile to AI progress. Like AI progress needs to keep happening at the same pace for NVIDIA to be going up again and again and again.

[Speaker 2] (1:47:30 - 1:48:01)

I mean, the earnings also to a certain degree seem to have caught up, right? There are like 40 times one year forward earnings, which is not ridiculous like it once used to be. Yeah.

I tried to learn about NVIDIA, like, correct me if I'm wrong, but the big distinction between NVIDIA chips and the incumbents of five or 10 years ago was the fact that they did tasks sequentially and NVIDIA does many tasks at the same time. Is that the main difference?

[Speaker 1] (1:48:02 - 1:49:38)

That's one way to put it. Mainly NVIDIA specialized for graphics. Graphics is a lot of matrix multiplications.

That's just how the math works. Matrix multiplications is parallel computations. And interestingly, it's just a very interesting coincidence.

It's not designed to be this way. Neural networks are also a lot of matrix multiplications. So because they specialize matrix multiplies to be fast for graphics, that core set of primitives that they built ended up being extremely a great fit for AI, like neural nets.

If AI was not neural nets, then GPUs wouldn't have mattered. But AI happened to be just basically neural nets at scale. And so all the primitives they built, all the software stack they built ended up being the core foundational building blocks for neural networks too.

All the neural network training libraries were built around it. Now, it's so hard for someone else new to come and change it. The only one who's managed to do this, I would say, is Google.

They built their own chips. They built their own software around it called JAX. And then they built their own accelerated linear algebra library called XLA.

And then they have their own data centers too. So they're the only ones who managed to do everything full stack, completely independent of the NVIDIA library and NVIDIA chips. Everybody else had either one or two of the pieces, but not all of it.

[Speaker 2] (1:49:39 - 1:50:27)

Right. Also, what is India's role in all this? Like, say, like I said earlier, like, this is genuinely how I feel, you know, Gen Z uses a particular word, they say FOMO all the time, like, fear of missing out.

I face that on a daily basis, because I keep reading about AI. But it does feel to me like, you know, the match is happening in another geography. And I'm talking to the commentator's friend about what is happening or reading what he's saying on Twitter, or X.

But what can India do? Or what should it do? And you can be like, honest about this, like, because there's something we want to incorporate.

And we want young people to go out and try at least.

[Speaker 1] (1:50:30 - 1:50:34)

I've said this before, I think India should definitely train its own models.

[Speaker 2] (1:50:36 - 1:50:48)

But wouldn't we arrive at the same answers that the incumbent models are arriving at if the data is largely democratized, and our data is also part of the training pool?

[Speaker 1] (1:50:49 - 1:51:28)

It doesn't matter. I think we should still build our own models, because there's so much more work to do on the models to make them reason and think and be good at things they're not good at today and be more agentic and do tasks and stuff like that. And India should have its own like, deep seek like company that trains models and like competes not just on Indian languages, but on global benchmarks.

And that will inspire the next generation of like engineers to come and work in those companies and build out the future.

[Speaker 2] (1:51:29 - 1:51:52)

Outside of fundamental models. I'm guessing this requires serious hardware and a reasonable

amount of data centers, chips, models, somebody young do like, say, a 25 year old boy or girl sitting out of Bangalore or Chennai or Mumbai or Delhi? What do they do specifically, like today, with no resources?

[Speaker 1] (1:51:52 - 1:51:56)

I would hope like they can raise some venture funding and try to do something.

[Speaker 2] (1:51:56 - 1:52:01)

Let's assume they're able to raise a million dollars because AI is hot right now. Then?

[Speaker 1] (1:52:02 - 1:52:51)

Well, it's pretty hard to do something meaningful with a million dollars, but certainly doable. The way I would do it is I would build a product that's pretty interesting and new and get users, raise more money, get more users and raise a little more money and then start to build your own models. Start with post-training on top of open source models, then start to like look into pre-training too.

And then get into the data centers. Like it's a multi-stage process. That's what I would do if I could start small.

But if you're already established, if you're not like this 25 year old young person, if you're already somewhat established, you have a presence, you have a name in the field and able to attract investments of higher magnitude, then I think you can go for the more ambitious targets right away.

[Speaker 2] (1:52:51 - 1:53:05)

Is there any like nuanced low-hanging fruit that Indians are not taking advantage of? Who want to start off? I don't know, maybe language.

Maybe we have access to...

[Speaker 1] (1:53:05 - 1:54:07)

I think voice. Most of the AIs are pretty bad at Indian voices. The speech recognition and speech synthesis are not necessarily good.

That's a place where you can make a clear difference because it's not a high priority for the Western labs to make it work. And there are like so many dialects and languages and like I think Indians are also more mobile app users and so voice is a more natural form factor of interaction. So really having that amazing real-time AI voice synthesis with broadly like support for all the Indian languages, nailing the dialects and accents and grammar would be a big deal.

It's easier said than done. It's not as easy as collecting data. I have to do a lot of evals and training and like iterations.

But it's definitely something that will matter a lot for the Indian market more than anybody else.

[Speaker 2] (1:54:09 - 1:54:12)

Because you're an investor as well. Would you buy Nvidia stock today?

[Speaker 1] (1:54:13 - 1:54:32)

I have my exposure to it. So I'm not selling, I'm holding. And I think I believe in...

like basically everybody's going to try to build super intelligence and general intelligence. And I think even if RL is working, I think you need a lot of compute to do it.

[Speaker 2] (1:54:32 - 1:54:39)

Is SF petty? Like if you said something bad about Nvidia, would you get lesser chips when you needed them?

[Speaker 1] (1:54:41 - 1:54:46)

I've not done that. So I don't know. But I think not.

I hope not. Yeah.

[Speaker 2] (1:54:47 - 1:54:58)

Right. What about the Indian outsourcing giants? Think of Infosys, TCS, Wipro.

What happens there?

[Speaker 1] (1:54:58 - 1:55:00)

I think they are just going to use AIs.

[Speaker 2] (1:55:01 - 1:55:07)

And what happens to all the people who are there? And if AIs are able to replicate...

[Speaker 1] (1:55:07 - 1:55:09)

They're not going to hire as many people going forward.

[Speaker 2] (1:55:10 - 1:55:27)

But the use case for an American company outsourcing to the Indian company to begin with was if one were to assume cheaper cost of labor. And now maybe an agent does. Then what happens to these companies on the whole?

[Speaker 1] (1:55:30 - 1:56:22)

Certainly less. They'll have to charge less. Some of it is actually based on relationships.

I know these AIs can do some of these things, but I would still trust you guys to do it without any bugs or errors. And until AIs are at a point of reliability where you just have no arguments not to use them, I feel like humans will still trust other human businesses to do stuff for them. But they'll just push them to like, hey, now that AIs can use this, why do you guys need three months to get it done faster?

Why do you guys need to charge this much, charges lower? I think they're going to push more on those shorter term trends rather than saying, hey, I don't think we need you guys anymore.

[Speaker 2] (1:56:22 - 1:56:30)

It's interesting that you say you guys. Now, at this point in life, do you view yourself as an American or an Indian?

[Speaker 1] (1:56:30 - 1:56:34)

No, I don't mean like you guys and the Infosys guys.

[Speaker 2] (1:56:34 - 1:56:36)

No, I don't mean it in a bad way. I don't mean...

[Speaker 1] (1:56:36 - 1:56:38)

No, no, no. Let me be clear.

[Speaker 2] (1:56:38 - 1:56:39)

We live here and we live there.

[Speaker 1] (1:56:40 - 1:57:09)

Yeah, yeah. So for the previous statement, I want to say it was simply between like, what would the software vendor say to the software provider? I mean, you guys in that sort of way.

It could mean, whether there are companies like Infosys smaller scale that do it in America too, where they clearly say somebody wants to move from Databricks to Snowflake and AIs cannot do the code translation, a human firm is actually doing that for them. Right.

[Speaker 2] (1:57:10 - 1:57:29)

But as somebody who has never like, I want to ask you like a favor at the end of all this, but as somebody who has never lived in the West, if you live there for long enough, does it become like, would one be conflicted in what you associate with?

[Speaker 1] (1:57:30 - 1:58:26)

I mean, you certainly change as a person, like you're not the same person anymore. Obviously, you're having a different outlook towards life and the world in general, but you are like, rooting obviously for India to succeed. And I don't see zero sum game between India and America, actually.

American businesses benefit a lot from Indian users. Indian businesses benefit a lot from American technology. And so, there's certainly like a lot of positive sum games to be played here.

And so, I'm actually, it's like one of those rare combinations that end up behaving this way. Not every country in the world is like, super friendly or like non-competitive with America.

[Speaker 2] (1:58:26 - 1:58:27)

Yeah.

[Speaker 1] (1:58:27 - 1:58:30)

And India is like, pretty lucky to be in this position.

[Speaker 2] (1:58:31 - 1:59:15)

With AI changing so many sectors, and you know, it's kind of like, re-plotting the map. Like, our crowd is fully entrepreneurship oriented, right? Like, wannabe entrepreneur crowd, all of us.

Yeah. Is there a sector that has tailwinds amidst all this? I'm thinking, think anything.

I could start a restaurant, I could start a steel company, I could start a SaaS business, I could start a t-shirt brand, I could start anything. Like, is there a sector with tailwinds where I will be served well in attempting entrepreneurship in the next decade?

[Speaker 1] (1:59:17 - 2:02:45)

I think there's going to be a lot of personalized apps built. Right now, if you want an app to work for you, what do you do is you go and file like, customer support bugs or like, you add them on Twitter and say, hey, this is not good, I want this, I want that. And what the app developer usually does is like, they have their own roadmap and they see the customer feedback, they look at the dominant feedback, and then they try to prioritize that into their roadmap.

But that feels inefficient. And in a world where AI can just write any software, I can build my own software. I can have my own kind of fitness app that'll work for my needs.

It'll know what I don't like working, what I don't like doing, what kind of workouts I like, how I feel that day, and I can program it to work for me. Same thing with health, same thing with like, tutoring, like I can have my own personal tutor for any topic. Maybe I don't know anything about finance, and maybe I want to get up to speed and like, I can tell it precisely.

And you know, I could try that with perplexity chat GPT-2, but then what if it doesn't actually tell it in the way I want, and I want to be able to build my own app for me. I think that layer is still not taken off, but it's certainly something that's waiting to happen, because as you can clearly see, software creation is getting a lot easier. So someone's going to be able to be that platform for deploying all these things in a secure way, and then people sharing apps that they've built for themselves with others.

And like, that's a social layer around it too. And I don't think anyone's really cracked this, and this might be a huge, huge market by itself. And I don't know how monetization is going to exactly work here.

Is it like micropayments, where if I use the app you created, I pay you? I don't know. Or is it going to be more traditional like ads, where different people are advertising to each other?

It's not clear. But what is clear is people are going to create a lot of personal stuff for them or the group of friends. Imagine I just wanted an app to split my payments with friends.

Earlier, you would go and use Splitwise, but what happened before Splitwise, you would do it all manually. Now I can just create a Splitwise app that's more custom, and I don't have to be like, oh, Splitwise doesn't have this feature. What if I can just directly build a better version of Venmo or something like that?

I think these are the kinds of things that I feel, even within businesses and enterprises, if I want to track the vacations people are taking, someone else needs to have built a vacation tracker or SaaS app. And now it's not needed, I can just build it myself. So all this stuff is going to change a lot.

And we're not yet there at this moment because there are still bugs, there are still things to fix. How do you deploy the app? Okay, cloud can write the code for you, but you have to actually deploy it.

You have to actually be in charge of where the data is living, all that stuff. But someone's going to abstract all these details out for you. It's going to feel super seamless.

And I think that's, in my opinion, this is the thing that will take off very quickly, but it's quite not

there yet.

[Speaker 2] (2:02:45 - 2:02:58)

Can you name a couple of apps that, as somebody who doesn't understand technology too much, such as me, that I have to use to get better? A better business, better, more efficient as a person?

[Speaker 1] (2:03:01 - 2:03:04)

I mean, I love to say perplexity, but...

[Speaker 2] (2:03:04 - 2:03:05)

I use perplexity already. Okay.

[Speaker 1] (2:03:07 - 2:03:22)

I think you should definitely give a shot at Cursor. What does Cursor do? Cursor is a coding assistant.

It helps you write code with an AI.

[Speaker 2] (2:03:22 - 2:03:24)

Even if I know nothing about writing code?

[Speaker 1] (2:03:25 - 2:04:11)

Right. You can just go and ask it to say, hey, I want to build a website with so-and-so, generate the code for me. But if you're like, I don't even want to be in charge of deploying it.

I think there's this thing called Replit or Bolt, where you can just go and describe an app you want to build, and the agent will build and deploy it for you. And I think that's where things are heading to. Bolt.

Bolt, B-O-L-T, or Replit, R-E-P-L-I-T. And sure, it's not going to work perfectly, but I feel like this is where things are headed, where I can just... You don't have to be a software engineer anymore to build an app.

And that's really...

[Speaker 2] (2:04:11 - 2:04:15)

But do you need a little bit of coding or a little bit of maths, maybe?

[Speaker 1] (2:04:15 - 2:04:15)

No.

[Speaker 2] (2:04:16 - 2:04:25)

No? But will I be able to produce an app via this method, which is as good as somebody who is a software engineer?

[Speaker 1] (2:04:26 - 2:04:39)

Not today. Right. As good as maybe like a lower level or like lower tier software engineer, yes.

Not as good as like the good ones or the best ones.

[Speaker 2] (2:04:39 - 2:04:45)

So if I were to have a kid, I shouldn't send him to an engineering college to study coding?

[Speaker 1] (2:04:46 - 2:05:22)

I think it still helps to be very good at infrastructure, back end, data centers, like floating point arithmetic, storage, all the core fundamentals are not going away. In fact, like I would say they're very essential in a world where AIs are taking care of the front end and the UI and all that stuff, because you have to know where the data lives. You have to know like how it is stored.

You have to know how it's deployed. And you have to know if a system goes down, how to fix it, debug it. Those things are still useful.

[Speaker 2] (2:05:23 - 2:05:35)

And last one or two questions. What is the future? If you were to like predict the next five years, you must have thought of this.

[Speaker 1] (2:05:36 - 2:07:50)

Yeah. I think we'll all have like a personal assistant. It's going to feel really amazing.

It's not going to be a luxury thing anymore. It's not just a thing billionaires had access to. It's going to feel like an iPhone where the same phone that the president of the US uses, you're going to be able to use too.

And by that, I mean, it's going to be pretty affordable. And that's going to make life a lot easier. And people are going to be able to build personalized things for them.

And that's been a lot more creative expression. Like whatever you want to exist in the world, you can make it happen. Not everyone in the world earlier used to be able to make something happen

when they wanted to.

They would use other people's creations. I think that's going to change. And that's going to feel very utopian.

That's the nice part of it. The dystopian part of it is, unfortunately, in the short term, there's going to be a lot of labor displacement. Not as many people are needed to get a work done anymore.

And so how people upskill themselves and adapt, those who are using AIs are definitely going to be well-positioned. So all that stuff is going to take place and how people react to it. That's already like, you don't need to build 10,000 people companies to be a trillion dollar company anymore.

So definitely, where are the next generation of graduates getting jobs? Existing big techs are laying off people or not hiring more. So all this stuff is definitely going to impact the market.

And it's very interesting that simultaneously by creating new value and making software creation easier, we're also displacing existing labor and value. So how people deal with all this is going to be interesting to watch. And I don't think anyone really knows how it will all play out.

[Speaker 2] (2:07:51 - 2:08:08)

Will the world be more complicated if a lot of this power access and determining the path forward is the decision making lies upon one or two geographies like is playing out today?

[Speaker 1] (2:08:09 - 2:08:39)

I think the technologies will be broadly accessible and secrets are not going to be lying in one or two places. And open source will ensure sufficient distillation to the rest of the world. I think what won't be democratized is access to compute, mainly because it takes a lot of money.

And that really depends on which countries choose to invest early on and later on in the process.

[Speaker 2] (2:08:39 - 2:09:18)

Right. I don't know what to ask you. I have in my notes that I should ask you about regulation and the future of that.

I don't know how to. I've read a fair amount about this and how a lot of people think that the incumbent AI players are trying to use it as a moat almost and capture the regulatory thing. Like, do you have any view on this?

Like, how should regulation? Let's say the government of India is listening or watching this show. What would be the right way for them to regulate AI?

And then B, what is the right way for America to regulate AI?

[Speaker 1] (2:09:22 - 2:11:35)

I mean, I think like regulating models is not necessarily a great idea. And it's not going to work in practice either. People are still going to be able to download a model and use it.

I think the best way is to regulate applications. Like, personally, what I feel is pretty concerning at this point is probably people using chatbots when they're kids and developing relationships with them and feeling suicidal when they don't get to enjoy the chatbots anymore or they don't respond in the way they want to. And kind of like taking your loneliness out on like an AI chatbot.

All that stuff is pretty, I find it concerning. Like maybe some people don't and they don't care and they just think this is not any different from how the internet used to be. But I think it is different.

So thinking about that application and like how do we make sure AI usage by kids is done on apps that are productive and useful and knowledge enhancing rather than feeling too companionship-like is worth thinking about. I don't think other stuff is worth regulating as much today. And we're kind of still very early in AI that moving slow is going to cost us a lot long-term.

And lot means like hundreds of billions or trillions of dollars. So it's best to keep accelerating right now and be mindful of like use cases like what I described that are clearly like dangerous, but otherwise like be pretty open-minded and build stuff and see how things play out. And I don't have a different answer to America or India.

[Speaker 2] (2:11:36 - 2:12:16)

I think it's the same answer Will the world get to a point as it gets more complicated that we all try and own our data a bit more where like, let's assume a model today is creating data from across the internet. Will the world go in a direction where Indians own Indian data, maybe like another country owns their data, and every model has to pay a fee to use that data as an input to train their models? In the sense, will things move behind a paywall or even if they don't move behind a paywall, will there be a fee?

[Speaker 1] (2:12:18 - 2:12:49)

It's possible. I think like in general, the internet has been global and fair use so far. I don't expect it to change.

I think if there are some tokens that are pretty valuable, and then people might want some kind of like token payment for it. It probably won't be on the internet. That's my guess.

[Speaker 2] (2:12:50 - 2:13:14)

Don't you find like that's happening already? More of it? Like right now, I find so much on the internet, which appears interesting, but it's behind a paywall.

But the question also will be that me as an individual, if I consume behind a paywall, a model which is then in turn going to distribute what is behind a paywall, should they be the same fee or should they be different fees?

[Speaker 1] (2:13:14 - 2:13:36)

I genuinely don't know. Because like the models are definitely like training on the content. So those who are training foundation models, they're not just like consuming the content once.

They're actually like distilling it, so they never have to consume it again. So it's a different kind of consumption to a human just reading an article.

[Speaker 2] (2:13:37 - 2:13:41)

Right. But even when I read an article, I consume it once.

[Speaker 1] (2:13:41 - 2:13:45)

Yeah, but like your memory and the model's memory are not comparable.

[Speaker 2] (2:13:46 - 2:13:48)

Right. But I'm not distributing it.

[Speaker 1] (2:13:51 - 2:14:20)

You might share the article with someone else, like say, hey, do you read this news? So you're attributing to it or you're going to use the wisdom you learned from it in some manner. I mean, perplexity, that's why we attribute it to a source.

Like we don't like say it's our content. And that way we give credit to the source and we're not actually training on the data. But ChatsGPT is different.

They actually train on all the data.

[Speaker 2] (2:14:21 - 2:14:38)

Right. OK, last question, Arvind. Because I'm feeling so left out in all of this, do you think it might be possible for me to come be an intern, work for maybe three months at Perplexity free of charge?

[Speaker 1] (2:14:39 - 2:14:43)

Well, you're way more accomplished for doing that.

[Speaker 2] (2:14:43 - 2:14:55)

But I'd love to. Like, this is genuine. Like, I feel like I'd love to come live there for a couple of months, learn some stuff and come back.

Because I do feel like I'm not learning enough right now.

[Speaker 1] (2:14:56 - 2:15:01)

I mean, we'd be very honored to have you. And I think I'm not joking.

[Speaker 2] (2:15:02 - 2:15:04)

I'm just going to be there in the next 30 days, maybe.

[Speaker 1] (2:15:06 - 2:15:06)

Sure.

[Speaker 2] (2:15:06 - 2:15:07)

We'd love to host you.

[Speaker 1] (2:15:10 - 2:15:55)

I guess I'd just say I love the spirit of how you're like having this learner's mindset. I think it's very inspiring and refreshing. So I don't think there's a lot you're missing out on.

The Internet has pretty much everything out there. And the world is running super fast that physical access matters way less anymore. I think it's more the amount of time you get to spend yourself with an AI model, using these apps, understanding where they fail and talking to the best people.

But interestingly, X has all of them literally talking all the time in real time. It's pretty nuts.

[Speaker 2] (2:15:55 - 2:16:07)

So it's not so much as learning from the model, but being around people who know what they're learning or who are learning what should be learned.

[Speaker 1] (2:16:07 - 2:16:14)

Yeah, definitely. It'll be like, like very refreshing to spend time and get a sense of the feel.

[Speaker 2] (2:16:15 - 2:16:26)

Super. But thank you for doing this. And you're going to be in India soon.

So if I'm not there, I'm going to host you when you're here in India. Yeah. Done.